# Mechanical waves and distant effects



**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use travel-time reasoning while distinguishing waves from the motion of material.

**SUCCESS CRITERIA - I can...**

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “medium” and “wave” to compare your model or result. Sentence starters are optional.

**medium**: the material through which a mechanical wave travels

**wave**: a disturbance that can carry energy and information

**signal**: a disturbance that carries information

## 01 Warm-up: bring back an earlier idea

A converging lens has f=15 cm and $d_o$=6 cm.
**Use** $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$.
**Find** $d_i$, magnification and image kind.

**Word help:** lens = a transparent element that changes light-ray directions

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

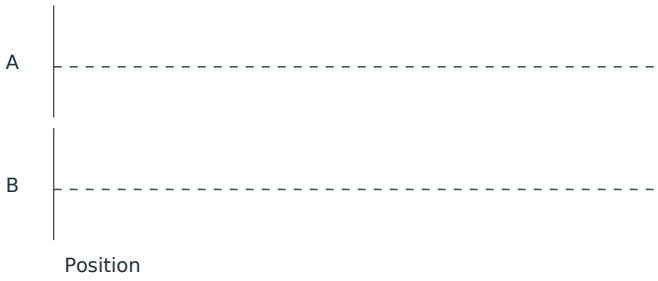
## 02 Worked example: explain the model

**Reference:** $d = v\,\Delta t$; $\Delta t = d/v$.
Use the supplied wave speed for its medium and wave type. These classroom models are not real hazard forecasts.

**Worked reasoning:** In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

**Your turn: Draw** the two waves and two corresponding relative-intensity bars.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




**Marzano check - circle:** ① ② ③

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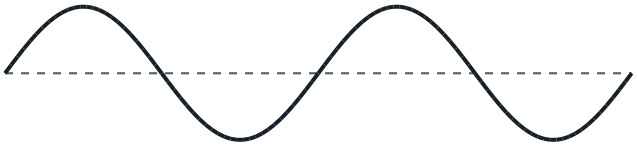
Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

**Draw** a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .



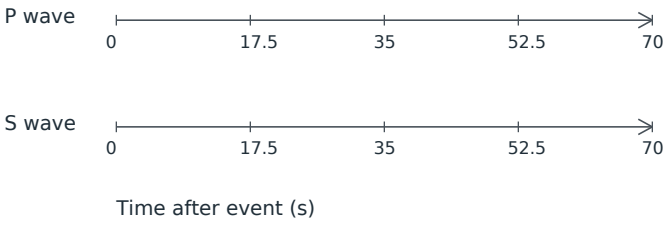
Sketch and label your response.

**Marzano check - circle:** 1 2 3

## 04 Apply the physics

A simplified seismic model uses P-wave speed 6 km/s and S-wave speed 3.5 km/s. A station is 210 km from the source.
**Calculate** each arrival time and the S-P delay.
**Mark** both arrivals on the timeline.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

Why does knowing wave arrival time alone not establish local damage?
**Name** other evidence.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

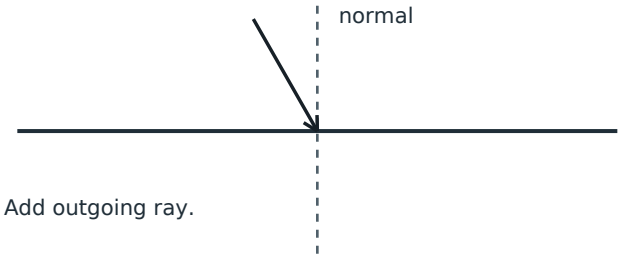
**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Light moves from a transparent medium with n=1.7 into air (n=1).
**Find** critical angle using sin $\theta_c$=1/n.
**Describe** incidence above it in the ideal model.
**Label** the media, normal and ray directions.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 07 Exit check

A shallow-water model uses v=√(gh). With g=10 and h=100 m, find speed and time for 10 km.
**State** a limitation.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .



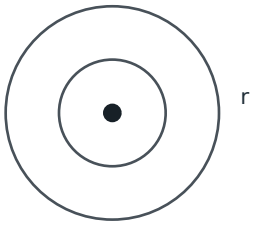

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A signal has intensity 8 W/m² at 10 m from an ideal point source.
**Find** intensity at 30 m and total radiated power.
**Explain** one limitation for ground waves.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .




Label distances and compare intensity.

**Marzano check - circle:** 1 2 3